

JOUR 8500 RLA

Experimental evidence for tipping points in social convention

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Centola, D., Becker, J., Brackbill, D., & Baronchelli, A. (2018). Experimental evidence for tipping points in social convention. *Science*, 360(6393), 1116-1119.

Background

- **Social conventions & theories**



Challenges:

classical equilibrium
stability analysis

Could be explained by:

Critical Mass Theory
(with a tipping point of changing)

Background

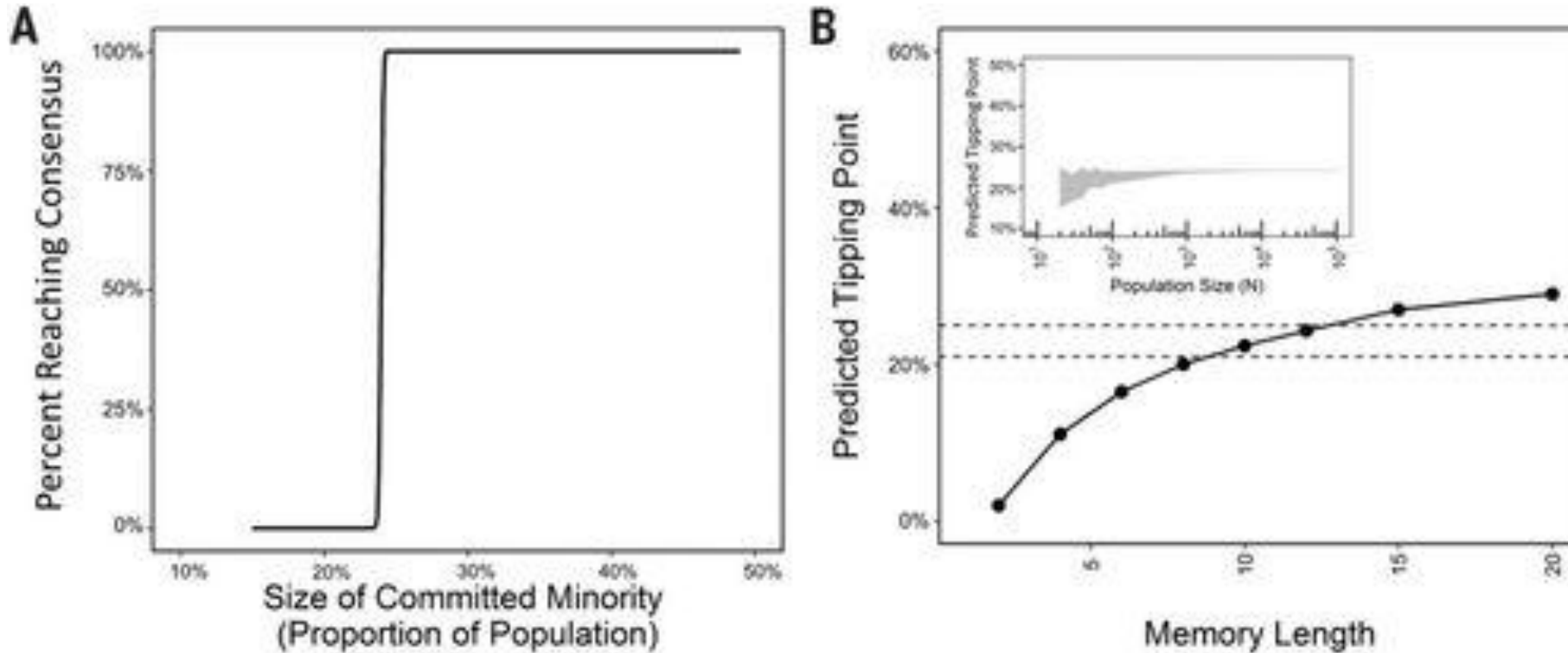
- **Critical Mass Theory:**

- Small groups of regular individuals can successfully initiate a change in social conventions

- **Evidence:**

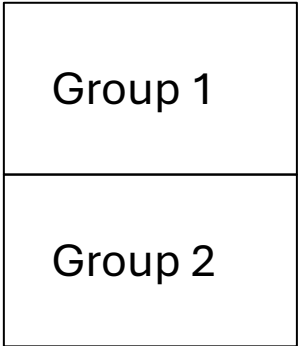
- Linguistic conventions: 10%
- Corporate leadership gender conventions: 30%
- Gender conventions: 40%

Experiment



- **Model:**
 - Parameters: individual memory length (M) and population size (N)
 - $M \rightarrow$ Size (reach the tipping point)

Experiment



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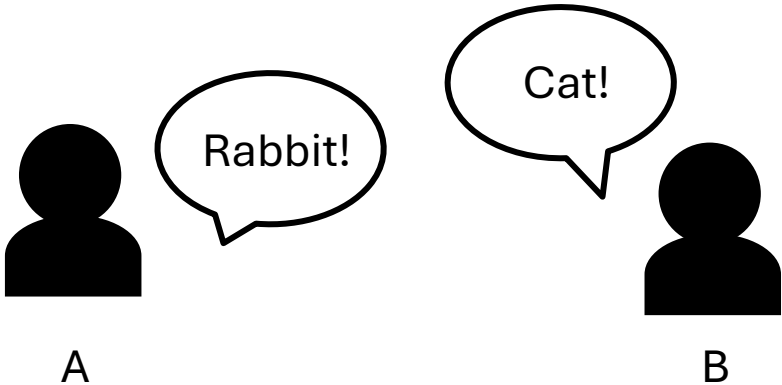


20-30 participants in each group (N = 194)

1) Within each group:

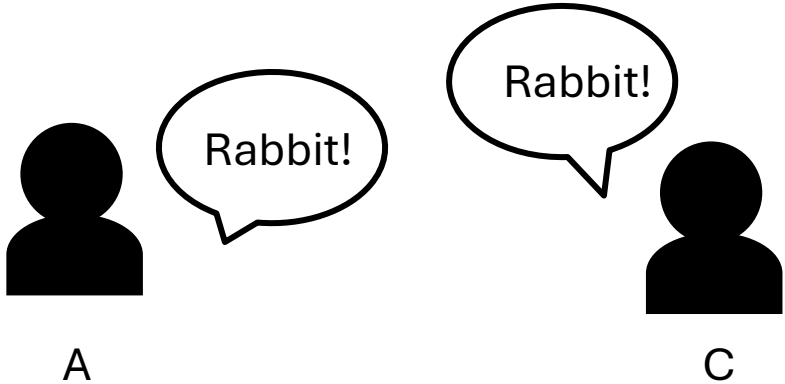


2) Randomly matched into pairs:

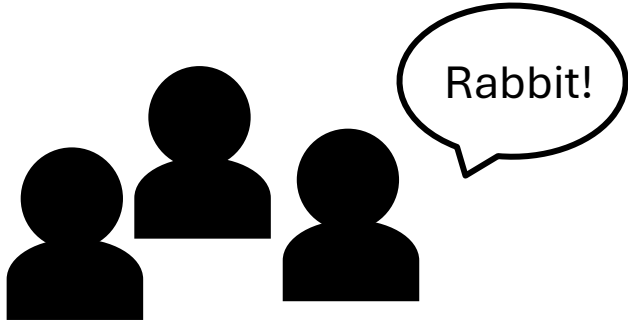


3) Coordinate a same name (succeed or fail)

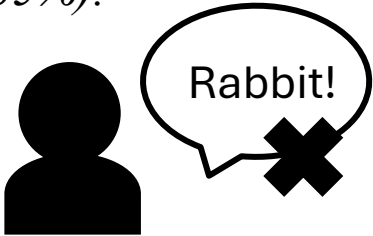
4) Interact with another member:



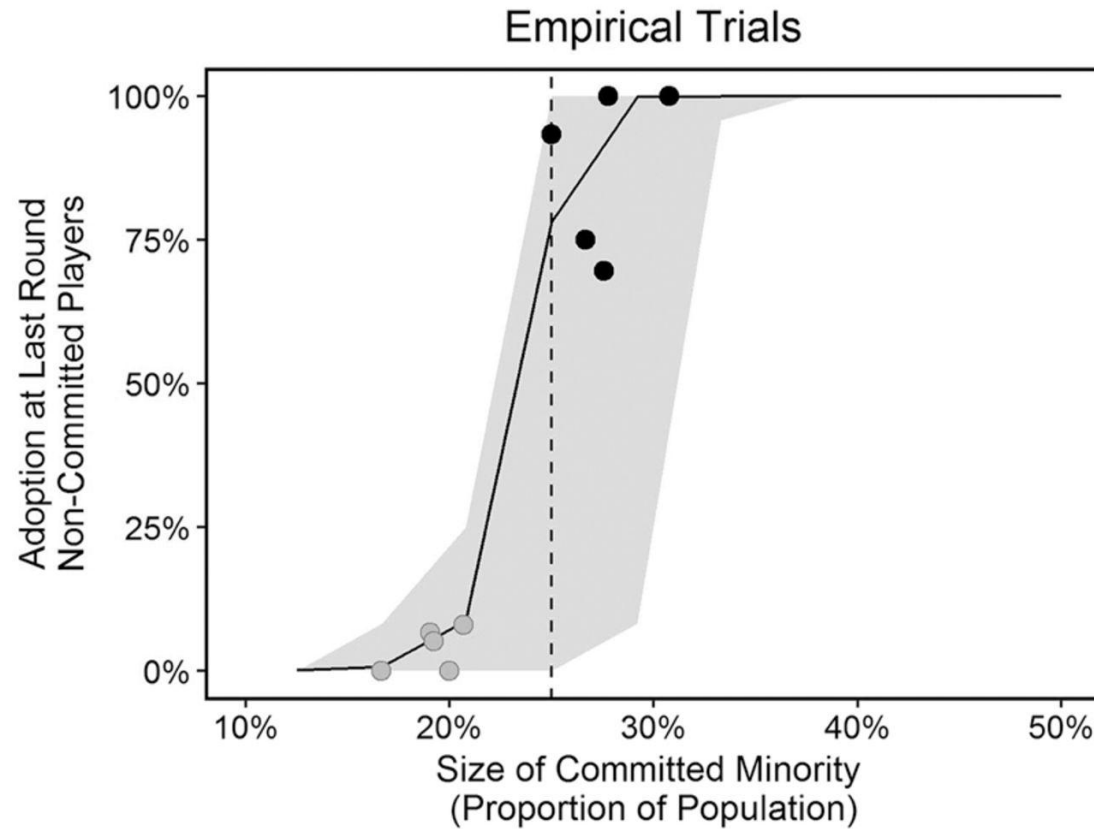
5) Convention established



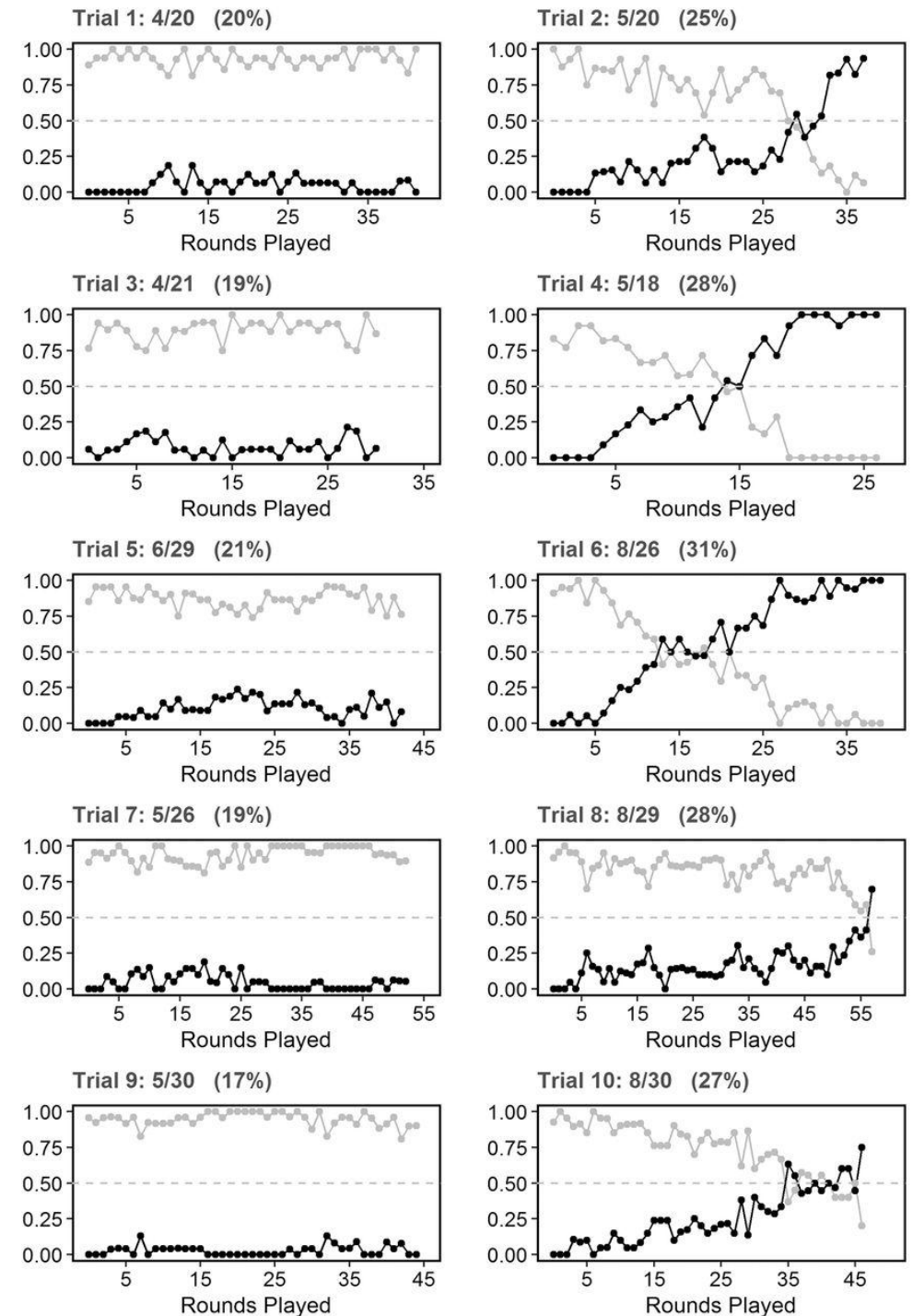
6) Introduced committed minority (15% - 35%):



Results



- Committed minorities (25%-27%)
→ uptake levels (72%-100%)
- Gender conventions (30%)



Questions

- **Generalization:**
 - What are the limitations of generalizing a model calibrated on small, artificially constructed groups? (Anastasia)
 - Are they trying to explain a very macro social phenomena (changes in social conventions) through a very well-crafted small experiment? (Jikai)

Questions

- **Generalization:**
 - This paper **lacks experimental settings in line with reality**, only considering the core variables that are being explored and simplifying other possible factors as much as possible. Is this also a common experimental design in social sciences?
(Jiarou)